

-
-
-
-
-

Chapter 6

Production Performance Reports



Production Performance Reports

Introduction

A number of useful performance measurement tools are provided which allow the user the flexibility to request performance measures over any range of dates.

User-defined Activity definitions and the underlying JBA Reporting Types allow users to configure activities and events to be reported and to tailor which data elements will be required in reporting each Activity. This flexibility supports Activity Based Costing (ABC) allocation information within the transaction database. Many of the performance reports can be used without modification to summarise overhead and resource cost allocations for ABC environments.

All reports include a printed heading that lists the selection parameter values used when the report was requested.

The reports are summarily grouped on the menu by:

- Item
- Work station
- Labour
- Scrap & rework

Item/Operation Efficiency Report

1... **1/MJR**
2...

This report calculates the average efficiency of each work station, for each operation for the items selected, within the selected date range. Efficiency is calculated as actual hours to standard hours (for the quantity produced by operation). Item totals are provided.



When Item/Operation Efficiency is selected on the Production Performance Reports menu, the Item Operation Efficiency Report Selection screen is displayed.

MJ670	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:46:16
Item Operation Efficiency Report Selection			
Organisational Model...? ☐			
WIP Locations from....? to			
Item Range from.....? to			
Shift Number.....			
Date Range from..... <u>8/12/95</u> to <u>8/12/95</u>			
Report Basis..... <u>2</u> 1=Machine, 2=Labour			
Include Rework..... <u>1</u> 1=Yes			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 SF MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From /To

This sets the limits for WIP locations to be included on the report.

Item Range From/To

This sets the limit for items to be included on the report. If entered, items must exist. Blanks indicate the inclusion of all items.

Shift Number

The report can be restricted to a single shift number. Only bookings made against this shift in the date range will be extracted for inclusion on the report. If the field is left blank, all shifts will be extracted.

Date Range From/To

These are mandatory fields. The report is produced using bookings that have been made for the date range selected. Both dates are inclusive. The default range restricts the report to the current day, but this can be extended.

Report Basis

Either 2 (labour) or 1 (machine efficiency) can be selected. This controls whether labour or machine bookings will be extracted for the work stations included in the report.

Include Rework

This indicates whether or not rework activity should be included on the report.

Enter 0 to exclude rework. Enter 1 to include rework.



Tip: If rework is to be included then the report program will also extract bookings for all activities that have the Reporting Type of Rework. This will have the effect of increasing the total time taken to produce the same quantity of product and will decrease the efficiency of the work station compared with the report that excludes rework.



Tip: By printing both reports (with and without rework) for the same work stations and the same date range you will be able to see the effect of rework on efficiency. This will help in the investigation of inefficiencies and problem solving. To obtain an analysis of the different reasons for rework, run the Rework Activity Code Analysis Report for the same work stations.



Press **Enter** to submit the job.

Material Usage Variance Report

1...
2...

2/MJR

This report presents the variances between standard and actual component material usage based on the quantities of parent items produced. Totals are reported for each parent item/operation. Variances are reported within WIP location.



When Material Usage Variance is selected on the Production Performance Reports menu, the Material Usage Variance Report Selection screen is displayed.

MJ705	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:46:33
Material Usage Variance Report Selection			
Organisational Model...? █			
WIP Locations From....? __ to __			
Consumed Items from...? _____ to _____			
Date range from..... 8/12/95 to 8/12/95			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 S2 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

A range of WIP locations may be selected.

Consumed Items From/To

A range of items may be entered. Blank indicates all items.

Date Range From/To

A range of dates may be entered. Blank indicates all items.

Inventory Cost Analysis Report

1...
2... **3/MJR**

This report prints the standard costs of production items received into stock using the appropriate cost route for that stockroom / item combination, compared with the earned standard costs derived from good quantity product bookings made against work stations over the date range specified.

The costs are detailed by cost elements and report cost element variances and totals.

The report is divided into two sections for each item. The first represents the 'this level' costs for the finished product and any purchased items contained within. The second section details the 'lower level' costs associated with any sub-assemblies that may be part of the finished item.

Both sections have the costing elements displayed from the standard values, the values as recorded after production and the variance. The 'this level' and 'lower level' costs are shown separately on the report.



When Inventory Cost Analysis is selected on the Production Performance Reports menu, the Inventory Cost Analysis Report Selection screen is displayed.

MJ700	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95
			17:46:47
Inventory Cost Analysis Report Selection			
From Department...? █		To Department...? _____	
From Item.....? _____		To Item.....? _____	
From Date..... 8/12/95		To Date..... 8/12/95	
Cost By..... 0		0 = Item Within Department 1 = Item	
F3=Exit F4=Prompt F8=Submit Job F18=Cost Presentation			
06-28 Sf MW KS IM II KENB20A PET166S2			

Fields:

From/To Department

Enter department names in the From Department and To Department fields, or leave blank to include all departments.

From/To Item

Enter item names in the From Item and To Item fields, or leave blank to include all items.

From/To Date

Enter dates in the From Date and To Date fields, or leave blank to include all dates. The system defaults to the current system date in both fields.

Cost By

This option allows sequencing of the report.

0 - To print by item within department.

1 - To print by item only. No department totals will be produced.

Functions:

F18

Cost Presentation. To amend the cost element presentation if required.

Work Station Efficiency Report



11/MJR

This report provides an analysis of how well production resources are performing against the standards you have defined. In addition, the report provides information regarding the costs of efficiency variances. The report can be produced for any time period (providing the work station records have been retained for the selected dates). Information can be restricted to a single shift in the period.

Work station efficiency can be analysed in terms of labour or machine performance, according to the type of information captured in the system. Rework can be included or excluded from the report. This enables the report to reflect the basic production rate efficiency providing the ability to analyse the percentage of Right First Time and to track improvements in quality over time.



When Work Station Efficiency is selected on the Production Performance Reports menu, the Work Station Efficiency Report Selection screen is displayed.

MJ600	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:47:02
Work Station Efficiency Report Selection			
Organisational Model...? █			
WIP Locations From....? __ to __			
Work Stations From....? ____ to ____			
Shift Number..... __			
Date range from..... 8/12/95 to 8/12/95			
Report Basis..... 2 1=Machine, 2=Labour			
Include Rework..... 1 1=Yes			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 82 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Shift Number

The report can be restricted to a single shift number. Only bookings made against this shift in the date range will be extracted for inclusion on the report. If the field is left blank, all shifts will be extracted.

Date From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive.

Report Basis

Either 2 (labour) or 1 (machine efficiency) can be selected. This controls whether labour or machine bookings will be extracted for the work stations included in the report.

Include Rework

Indicates whether or not rework activity should be included on the report.

Either 0 to exclude rework, or 1 to include rework.



Tip: If rework is to be included then the report program will extract bookings for all activities that have the Reporting Type of Rework. This will have the effect of increasing the total time taken to produce the same quantity of product and will decrease the efficiency of the work station compared with the report that excludes rework.



Tip: By printing both reports (with and without rework) for the same work stations and the same date range you will be able to see the effect of rework on efficiency. This will help in the investigation of inefficiencies and problem solving. To obtain an analysis of the different reasons for rework, run the *Rework Activity Code Analysis Report* for the same work stations.



Press **Enter** to submit job.

Work Station Utilisation Report

1... **12/MJR**
2...

This report compares the total time that a machine was available for work with the amount of time that it was actually being used within a selected date range.

Non-usage of production resources will usually represent overhead costs that are not directly related to the goods normally produced using them. This report will provide information enabling you to identify the areas where these overheads occur, and the amount of under-utilisation which will help in deciding how to spread indirect costs within the organisation.

Work Station Utilisation can be analysed in terms of labour or machine performance, according to the type of information captured in the system.

Production will be detailed by item/operation produced. Time which has been reported as down time is not included in the calculations.



When Work Station Utilisation is selected on the Production Performance Reports menu, the Work Station Utilisation Report Selection screen is displayed.

MJ620	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:47:21
Work Station Utilisation Report Selection			
Organisational Model...? █			
WIP Locations From....? ____ to ____			
Work Stations From....? ____ to ____			
Date range from..... 8/12/95 to 8/12/95			
Item.....? _____ (Blank for all)			
Utilisation Basis..... 2 1=Machine, 2=Labour			
F3=Exit F4=Prompt F8=Submit Job			
05-35 84 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Work Stations From/To

This sets the limit for work stations to be included on the report.

Date Range From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive.

Item

If you want the report to compare work station availability with the amount of capacity consumed in the production of a single item, enter an item code in this field. The bookings made against the item in the date range will be included and the total printed as a proportion of the total work station capacity.

Utilisation Basis

The option is given to report on labour or machine utilisation.

- 1 - For machine utilisation. The system will compare standard machine hours to actual hours booked to work stations.
- 2 - For labour utilisation. The system will compare standard labour hours to actual hours booked to operators or crews.

Set Up Cost Report

1... **2...** **13/MJR**

Bookings of set-up costs can be made against item/operations. This report will show the actual set-up costs booked. Labour and machine hours as well as labour, machine and material costs are presented.



When Set Up Cost Report is selected on the Production Performance Reports menu, the Set Up Cost Report Selection screen is displayed.

MJ690	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:47:36
Set-up Cost Report Selection			
Organisational Model..? _			
WIP Locations From....? _ to _			
Items from.....? _____ to _____			
Date range from..... 8/12/95 to 8/12/95			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 S2 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Mode

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Items From/To

This sets the limit for items to be included on the report. If entered, items must exist. Blanks indicate no limit.

Date Range From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive.

Direct Overhead Variance Report

1...
2...

14/MJR

The direct overhead variance report displays the difference between the actual variable overheads from the quantity produced at operations, with the standard variable overheads.



When Direct Overhead Variance is selected on the Production Performance Reports menu, the Direct Overhead Variance Report Select screen is displayed.

MJ612	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:47:51
-------	-------------------------------	--------------	---------------------

Direct Overhead Variance Report Select

Organisational Model...? __

WIP Locations From....? __ to __

Work Stations From....? ____ to ____

Items From.....? _____ to _____

Date From..... 8/12/95 to 8/12/95

Include Rework..... @ 1=Yes

F3=Exit F4=Prompt F8=Submit Job

05-35 **SA** MW KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

Set the limits for WIP locations to be included on the report.

Work Stations From/To

Set the limit for work stations to be included on the report.

Items From/To

This sets the limit for items to be included on the report. A range of items may be entered. Blank indicates all.

Date From/To

The report uses bookings made for the date range selected. Both dates are inclusive. Blank indicates lack of limit.

Include Rework

Enter 0 to exclude rework activity from the report. Enter 1 to include rework activity on the report.

Down Time Report

1...
2...

15/MJR

This report provides an analysis of periods when work stations have not been operational. The report provides analysis by down time reason code, period and work station. Down time can be booked against a work station by using the appropriate activity type code.



When Down Time Report is selected on the Production Performance Reports menu, the Work Station Downtime Report Selection screen is displayed.

MJ695	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:48:12
Work Station Downtime Report Selection			
Organisational Model...? █			
WIP Locations from....? __ to __			
Work Stations from....? ____ to ____			
Date Range from..... 8/12/95 to 8/12/95			
Reason Code.....? __ (Blank for all)			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 S2 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Work Stations From/To

This sets the limit for work stations to be included on the report.

Date Range From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive.

Reason Code

Leave blank.

Indirect Overhead Report



16/MJR

Bookings of indirect overheads (labour, machine and material) can be made against item/operations. This report will show the indirect bookings that have been attributed to item/operations using an appropriate activity type code.



When Indirect Overhead Report is selected on the Production Performance Reports menu, the Indirect Overhead Report Selection screen is displayed.

MJ685	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:48:35
Indirect Overhead Report Selection			
Organisational Model..? █			
WIP Locations From....? ____ to ____			
Items from.....? _____ to _____			
Date range from..... 8/12/95 to 8/12/95			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35	SA	MW	KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Items From/To

This sets the limit for items to be included on the report. If entered, items must exist. Blanks indicate no limit.

Date Range From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive.

Department Cost Analysis



17/MJR

This report compares the cost of produced inventory costed at the prevailing production standard with the actual booked costs, providing a fully costed efficiency variance.

This is the department efficiency variance by period - earned standard versus actual.



When Department Cost Analysis is selected on the Production Performance Reports menu, the Department Cost Performance Selection screen is displayed.

Fields:

From/To Department

Enter department names in the From Department and To Department fields, or leave blank to include all departments.

From/To Item

Enter item names in the From Item and To Item fields, or leave blank to include all items.

From/To Date

Enter dates in the From Date and To Date fields, or leave blank to include all dates. The system defaults to the current system date in both fields.

Cost By

This option allows sequencing of the report.

0 - Item Within Department. To print by item within department.

1 - Item. To print by item only. No department totals will be produced.

2 - Department. To printed by department only.

Transaction Detail

0 - No.

1 - Yes.

Functions:

F18

Cost Presentation. To amend the cost element presentation if required. The Cost Presentation window is displayed.

Direct Labour Variance Report



21/MJR

This report compares the actual direct labour expended for the produced quantity at operations with the standard direct labour for the quantity produced. Direct labour resulting from any booking that has an activity type associated with the direct labour reporting type that triggers direct labour transaction types.

The total labour variance is the difference between the standard labour cost for the actual production at an item/operation, and the actual labour cost. This is broken down into a rate variance and an efficiency variance. Variances can arise because of differences between the standard and actual rates paid for labour and because of differences between the standard and actual production quantities achieved for the hours used.

The rate variance is the difference between the standard labour rate and the actual rate for the actual number of labour hours booked. The actual labour rate printed is the average rate calculated from the bookings made within the date range.

The labour efficiency variance is the difference between the standard labour hours allowed for the quantity produced and the actual hours booked, valued at the standard labour rate.

Variance totals are printed for individual item/operations, and are accumulated by item for each work station in the model.



When Direct Labour Variance is selected on the Production Performance Reports menu, the Direct Labour Variance Report screen is displayed.

MJ610	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:48:56
Direct Labour Variance Report			
Organisational Model...? ☐			
WIP Locations From....? to			
Work Stations From.....? to			
Items From.....? to			
Date From..... 8/12/95 to 8/12/95			
Include Rework..... 1 1=Yes			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35		MW KS IM	II KENB20A PET166S2

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Work Stations From/To

This sets the limit for work stations to be included on the report.

Items From/To

This sets the limit for items to be included on the report. A range of items may be entered. Blank indicates all.

Date From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive. Blank indicates all.

Include Rework

Indicates whether or not rework activity should be included on the report. Enter 0 to exclude or 1 to include rework.



Tip: If rework is to be included then the report program will extract bookings for all activities that have the Reporting Type of Rework.



Tip: This will have the effect of increasing the total time taken to produce the same quantity of product and will increase variances (of the item/operations with rework) compared with the report that excludes rework.

By printing both reports (with and without rework) for the same work stations and the same date range you will be able to see the effect of rework on variance. This will help in the investigation of inefficiencies and problem solving. To obtain an analysis of the different reasons for rework, run the Rework Activity Code Analysis report for the same work stations.

- $\text{Actual Rate} = \text{Actual Cost} \div \text{Actual Hours}$
- $\text{Standard Rate} = \text{Standard Cost} \div \text{Standard Hours}$
- $\text{Rate Variance} = (\text{Standard Rate} - \text{Actual Rate}) \times \text{Actual Hours}$
- $\text{Performance Variance} = (\text{Standard Hours} - \text{Actual Hours}) \times \text{Standard Rate}$
- $\text{Total Variance} = \text{Rate Variance} + \text{Performance Variance}$ or $\text{Standard Cost} - \text{Actual Cost}$

Scrap and Rework Analysis Report



31/MJR

This report provides an analysis of the quantities and costs of scrap and rework produced by item/operations on work stations over a range of dates.



When Scrap And Rework Analysis is selected on the Production Performance Reports menu, the Scrap And Rework Analysis Selection screen is displayed.

MJ680 Z1 - GB Demonstration Company User GBJBAMS 8/12/95
17:49:15

Scrap and Rework Analysis Selection

Organisational Model.....? █

WIP Locations from.....? to

Items from.....? to

Date range from..... 8/12/95 to 8/12/95

F3=Exit F4=Prompt ENTER=Submit Job

05-41 █ MW KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report.

Items From/To

Enter an item range or leave blank for all items.

Date Range From/To

Enter a date range.

Scrap Reason Code Analysis Report

1...
2...

32/MJR

When scrap is reported, a reason code must be associated with it. This report displays a breakdown of scrap by reason code, all scrap costs incurred for the same reason, and the total scrapped quantity by reason, by item/operation, between the selected dates. Processes yielding abnormally high scrap can be identified.



When Scrap Reason Code Analysis is selected on the Production Performance Reports menu, the Scrap Reason Code Analysis Report Selection screen is displayed.

MJ715	Z1 - GB Demonstration Company	User GBJBAHS	8/12/95 17:49:32
-------	-------------------------------	--------------	---------------------

Scrap Reason Code Analysis Report Selection

Organisational Model.....? __

WIP Locations from.....? __ to __

Items from.....? _____ to _____

Date range from..... 8/12/95 to 8/12/95

Reason Code.....? __

F3=Exit F4=Prompt ENTER=Submit Job

05-38 **SF** MW KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

Enter the range of WIP locations to be included on the report. If blank, there is no limit. Only WIP locations that are in the selected organisational model will be included.

Items From/To

Enter the range of items to be included on the report. If blank, there is no limit. Only items that are in the selected organisational model will be included.

Date Range From/To

Enter the range of dates to be included on the report.

Reason Code

If entered, the report will relate only to this single reason code. If blank, all reason codes are included.

Rework Activity Code Analysis Report

1...
2...

33/MJR

This report provides a breakdown of rework by a rework activity code. Activity types may be defined for different types of rework in your organisation. All rework bookings have an associated activity. This enables the report to accumulate all rework costs for the same rework activity.

Rework activity costs are detailed by labour, machine, material, overhead 1 and overhead 2 cost elements by item/operator within the selected date range.



When Rework Activity Code Analysis is selected on the Production Performance Reports menu, the Rework Activity Code Analysis Report Selection screen is displayed.

MJ720	Z1 - GB Demonstration Company	User GBJBAMS	8/12/95 17:49:47
Rework Activity Code Analysis Report Selection			
Organisational Model.....? __			
WIP Locations from.....? __ to __			
Items from.....? _____ to _____			
Date range from..... 8/12/95 to 8/12/95			
Activity Code.....? _____			
F3=Exit F4=Prompt ENTER=Submit Job			
05-30 SF MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the live organisational model.

WIP Locations From/To

This sets the limits for WIP locations to be included on the report. Alternatively, you can leave the fields blank to include all locations.

Items From/To

This sets the limits for items to be included on the report.

Date Range From/To

The report is produced using bookings that have been made for the date range selected. Both dates are inclusive.

Activity Code

If the report is to be produced for a single rework activity, enter the required activity code. Only rework bookings carrying this activity code will be extracted for the report. If the field is left blank, all rework bookings will be used to produce the report, with each activity code detailed separately.